Engineering Mechanics 1 · Unit_04 · UNIT 4

DRAFT — UE26ME142A · Unit_04 · UNIT 4 · CLASS 56.1 Friction Numerical

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · CLASS 56.1 Friction Numerical
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

CLASS 56.1 Friction Numerical

1 CLASS 56.1 Friction Numerical

Study focus: CLASS 56.1 Friction_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000158 · EV0000159

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Question 6/33 Determine the distance S to which the 90 kg painter can climb without causing the 4 meter ladder to slip at its lower end A.

Key details

- The top of the 15 kg ladder has a small roller.
- And at the ground, the coefficient of static friction is 0.25, the mass center of the painter is directly above our feet.
- This is the figure we were supposed to follow for this particular question.
- 25 that is mu s is equal to 0.25 at the ground.
- Then at the B, this roller is resting on the vertical wall and we are going to neglect the frictional force at this roller because we will assume it is an ideal condition.
- She is trying to climb the ladder.
- When she is at a distance of s with respect to A, then she will be in a state of equilibrium.
- If she further climb up the ladder, then the ladder is going to slip.
- In this direction.
- We know the length of the ladder is 4 meter.
- I call this point as G.
- G is the center of gravity.
- of the ladder through which the weight of the ladder will be acting vertically downwards.
- W stands for weight of the ladder.
- That will be equal to mass of the ladder into G.
- The reactions here at A, since the ladder is directly resting on the ground, there will be normal reaction NA as well as there will be a frictional force.

- Since the ladder is having a tendency to slip in the left side direction, frictional force here will be right side here, A.
- Next, we know the coefficient of friction at A, so F A can be written as mu into N A.
- Next and B there will be a normal reaction from the wall onto the roller.
- So NB will be horizontal because the wall is vertical.
- Weight of the painter will be equal to mass of the painter, that is 19 to 9.81 newton.
- Then after that we can find the value of S.
- The first condition of equilibrium here is sigma f of x is equals to 0 considering the right side component of the forces positive left side negative.
- F is towards right side so plus e of a then n b is towards left so minus n b is equal to 0.
- These two are the only forces along X direction.
- What is FA here is 0.25 times the of NA.
- That will be equals to N B.
- R in other ways N B is equals to 0.25 times the of.
- And I'll call this as equation one.
- Na is upwards so plus Na, then weight of the painter is acting downwards, so minus WP, weight of the ladder is acting downwards, so minus W L is equal to 0.
- Here we know WB that is 90 into 9.81.
- We know WL that is 15 into 9.81.
- After substituting WB and W L here we will get the value of NE.
- That is 1030.5 newton.

Example or application from the lecture

- Here what is given is there is a ladder AB, the length of the ladder is 4 meter, then at A the ladder is directly resting on the ground, that's why they have given the coefficient of static friction between the ladder and the ground as point.
- There is a painter, her mass is given as 90 kg.
- I am going to consider the ladder for the free body diagram.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 05:00–09:59

We can substitute this value of NA in the equation 1 and we can find the value of N B.

Key details

- Next, the last condition of E helium is moment.
- Since at A we have two forces, I am going to apply moment at A as 0 so that I can apply those two forces in this equation.
- Moment of this W is.
- W P into distance.
- This distance I will call it as X1.
- Then how to find this distance X1?
- To find this distance X1 I should know this angle theta.

- Then how to find this angle theta?
- I'll correspond to C.
- Theta will be equals to I know the hypotenuse, I know the adjacent side, so it will be equal to cos inverse of adjacent is 1.5 by hypotenuse is 4.
- This is the value of theta.
- Moment because of this WP is clockwise, so minus WP.
- This is the force into perpendicular distances x1.
- And what is X1?
- X1 will be equal to S cos theta.
- From the small triangle here.
- Then come to this WL.
- W n is also acting in the clockwise direction.
- This distance is 2 and this angle is known to as theta, this will be equal to 2 cos theta.
- W n into 2 cos theta.
- This is the moment because of the weight of the ladder.
- Then it comes to this NB, it is acting in the counterclockwise direction.
- Since the line of action is horizontal, the distance will be vertical, that is this C D.
- Side C B.
- Then how to find the C B?
- We know C B will be equals to under root of 4 square minus 1 point five square because a C B is the right angle triangle.
- X1 we know S cos theta.
- Then minus weight of the ladder.
- Weight of the ladder is 15 into 9.81.
- That is equal to 1 for 14.
- one forty seven point one five into two cos theta is sixty seven point nine seven then plus N B N B we know N B is two fifty seven point 512 into C B.
- C B is under root of 4 square minus 1.5 square that is 3.708 will be equal to 0.
- In this except yes, we know all the other terms.
- If she further climbs up the ladder, then the ladder is going to slip down the plane.
- That is what the meaning.
- Next question 6 slash 36.
- The homogeneous square body is positioned as shown in the figure.
- If the coefficient of static friction at base power four determine the critical value of angle theta which will which slipping will below which slipping will occur.
- Then friction at A.

Example or application from the lecture

- To find the theta, I am going to consider this big triangle A, this is B.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

Core explanation · 10:00–14:59

Here what is given here is there is the homogeneous square body and that body is having a roller at A and the body is resting directly on this horizontal surface at B.

Key details

- The side A B of this body is making an angle of theta with respect to this horizontal surface.
- It is that value of theta for which the body will be in a state of equilibrium.
- If there is a if the value of this theta decreases, then the body is definitely going to slide or slip.
- Along this surface, it's going to move in this direction.
- Since the body is in direct contact with the surface, there will be a normal reaction as well as there will be a friction force.
- This body will have a tendency to slip in the right side direction, the frictional force here will be towards left side.
- I will call that as FB.
- FB will be equal to mu into normal reaction.4 N this year normal reaction is NB.
- Next, there will be.
- Reaction from this surface onto this roller that will be always making an angle of 90 degrees with the surface.
- Normal reaction NE.
- Normal reaction at A.
- For that I am going to first find out the geometric center of this.
- Then I am going to.
- Since it we don't know exactly what is this mass or weight, I'm going to call it this is just W.
- This is along its direction, so we no need to find the angle.
- This is along y direction, no need to find the angle.
- But here it is inclined.
- This is 60 degree.
- I am going to draw a horizontal line here.
- If this is 60, this angle will become 30 because the angle between this norm reaction and the surface is 90 degree.
- That's why this angle will become 30 degree.
- Then we will find the value of theta.
- X component of this Na will be towards right side.
- Then F B is towards left side.
- I can write here N B is equals to 2.165 times of NE.
- This is the relationship between N and we will call it as equation one.
- The upward component of N A will be Na sin thirty.
- Since it is upwards it's plus Na sin thirty and W is acting downwards so minus W then N B is acting upwards so plus N B is equals to 0.
- So N A will be equals to once N A will be equals to W divided by 2.664 because we know N B is 2.165 times zero in A.
- We will get this expression and call this is equation 2.

- Next, the last condition of equilibrium, moment about any point 0, I'm going to apply this at B because at B I have two forces.

Example or application from the lecture

- Let's start with the free body diagram of this homogeneous body.
- Next we have to consider the weight of this body, homogeneous body.
- Consider draw one line the distance of s by 2 then s by 2.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000166 · EV0000167 · EV0000168 · +1 source units in pack.json

Core explanation · 15:00–19:59

Two components, one is horizontal and one is vertical.

Key details

- This is NA cos thirty and this is NA sin 30.
- This is force.
- Into perpendicular distance we need to find.
- Then how to find the perpendicular distance.
- You can see there is a rectangle triangle here, it's like this.
- This side is this is A and this is B.
- A B we know it's the one of the side of a square, it will be S and this angle we know theta.
- Then if I call this point as C then this will be equal to S cos theta and this will be equals to S sine theta.
- That's why here for this particular force the perpendicular distance will be the height of this triangle that will be equal to s sin theta.
- S sine theta.
- Then come to this force it is acting in the clockwise direction again.
- Perpendicular distance will be the base of this channel that will be S cos theta.
- Next come to this W for W.
- You have to remember one thing.
- This distance that is also S by 2.
- Since we know the coordinates of this particular point G, that is the point to which.
- The mass of this body is acting vertically downward.
- Since this is S by 2 and S by 2, we know the coordinates of this G with respect to B.
- That's why resolve it along this line and normal to this line.
- For this, we should know the angle.
- The angle here will be connect.
- We have to look into this figure.
- This angle is theta.
- If this is theta, this angle will be also theta because mutually perpendicular.
- Cos theta this will be W sine theta.
- It is clockwise so minus W sine theta.
- perpendicular distance will be S by two.

- Then come to the component.
- This component it is counterclockwise, so plus W into cos theta into perpendicular distance will be s by 2.
- This will be equal to 0.
- This will be.
- We have sine theta here, we have sine theta here also.
- And we have cos theta here and we have cos theta here.
- Once we simplify this, we are going to get the expression like this.
- Sine t sine theta into point A to five will be equals to cos teta into point three one 23 then sin by cos will be equals to tan theta that will be equals to 0.3785 then theta will be equals to tan inverse of 0.3785 that is equals to twenty point seven three.
- This is the value of theta for which this homogeneous body will be in a state of test.
- If there is a decrease in the value of theta, then this is going to slide down the plane.
- That is what the meaning.
- That's it for today's class.

Example or application from the lecture

- Let's start with the horizontal component, it is acting in the clockwise direction.
- For that I am going to consider this right angle triangle.
- If you consider W as it is, then we don't know this perpendicular distance and we can't find it so easily.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 56.1 Friction_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.1_Friction__Numerical__816881a9ab73e7db.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 56.1 Friction_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.1_Friction__Numerical__816881a9ab73e7db.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 56.1 Friction_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.1_Friction__Numerical__816881a9ab73e7db.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
CLASS 56.1 Friction_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.1_Friction__Numerical__816881a9ab73e7db.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md have no readable `.slides.md` text; their original page images are included.
3. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
4. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
5. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
6. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
7. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

8. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00042	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00043	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00044	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00045	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

C Revision and retrieval practice